Sri Lanka Institute of Information Technology

Data Warehousing &
Business Intelligence- IT3021
Assignment 1

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Database	WideWorldImporters

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1.Introduction

In modern organizations, data plays a critical role in supporting strategic decision-making. However, data stored in operational systems is often not optimized for analytical queries. Data warehousing addresses this challenge by integrating and organizing data into a centralized repository designed for analysis and reporting.

This project aims to design and implement a data warehouse using the WideWorldImporters dataset. The system is structured using a dimensional modeling approach, specifically a star schema, to improve query performance and simplify data analysis. The project also includes the development of an ETL process to extract data from the source system, transform it into a suitable format, and load it into the data warehouse.

Furthermore, an OLAP cube is implemented to support multidimensional analysis, enabling users to perform complex queries and gain valuable business insights. This project demonstrates the complete workflow of building a data warehouse solution, from data extraction to analytical reporting.

2. Dataset Selection

2.1 Dataset Overview

The dataset selected for this project is WideWorldImporters Transactional database. This database is a sample OLTP (Online Transaction Processing) database provided by Microsoft. It contains real- world business data related to customers, products, suppliers, invoices and sales transactions. Which are representing daily business operations of the retail organization. So these reasons make WideWorldImporters database suitable for data warehouse and perform analytical processing.

2.2 Dataset Justification

The WideWorldImporters dataset was selected due to the following reasons:

- The dataset satisfies the minimum data volume requirement, as it contains a large number of transactional records (InvoiceLines-228 265), enabling meaningful aggregations, cube processing and reporting.

- The dataset uses multiple data source types, including:
  -SQL Server database tables (OLTP system)
  -CSV file (customers.csv)

  This satisfies the requirement for integrating heterogeneous data sources using ETL.

- The dataset provides rich dimensional attributes suitable for building a star schema, including,
  -Customer information (name, category, contact details)
  -Product information (brand, size, pricing)
  -Salesperson details
  -Time dimension (invoice dates)

- The dataset supports a Slowly Changing Dimension (SCD),
  -The Customer dimension can be identified as an SCD Type 2, where attributes such as customer category, location, contact details may change over time.
  -This is implemented using fields such as StartDate, EndDate and IsCurrent.

- The dataset supports hierarchical structures, which are essential for analytical queries and drill-down operations,
  -Geography hierarchy:
  Continent → Region → Country → State/Province → City

  -Time hierarchy:
  Year → Quarter → Month → Day

- The dataset is a true OLTP system, consisting of normalized transactional tables (Invoices and InvoiceLines) with foreign key relationships, rather than pre aggregated analytical data.

- The presence of fact level data (InvoiceLines) allows for detailed measure calculations such as,
  - -Sales Amount
  - -Profit
  - -Tax
  - -Quantity sold

- With the available data volume and structure, the dataset is suitable for,
  - -ETL implementation using SSIS
  - -Star schema design
  - -Cube processing and aggregation for Assignment 2
  - -Business reporting and visualization

Although the WideWorldImporters dataset is provided by Microsoft, it is not a pre-built data warehouse or OLAP dataset. It is a fully normalized OLTP system that requires complete data warehouse design, ETL development and dimensional modelling from scratch. Therefore, it satisfies the requirement of selecting an OLTP dataset and allows the implementation of all data warehousing concepts independently

2.3 Dataset Attributes and Summary

The selected dataset consists of the following key tables,

Source Type	Source	Row count
SQL Server Table	Sales.CustomerCategories	8
	Sales.Invoices	70 510
	Sales.InvoiceLines	228 265
	Warehouse.StockItems	227
	Application.People	1111
	Application.Cities	37 940
	Application.StateProvinces	53
	Application.Countries	190

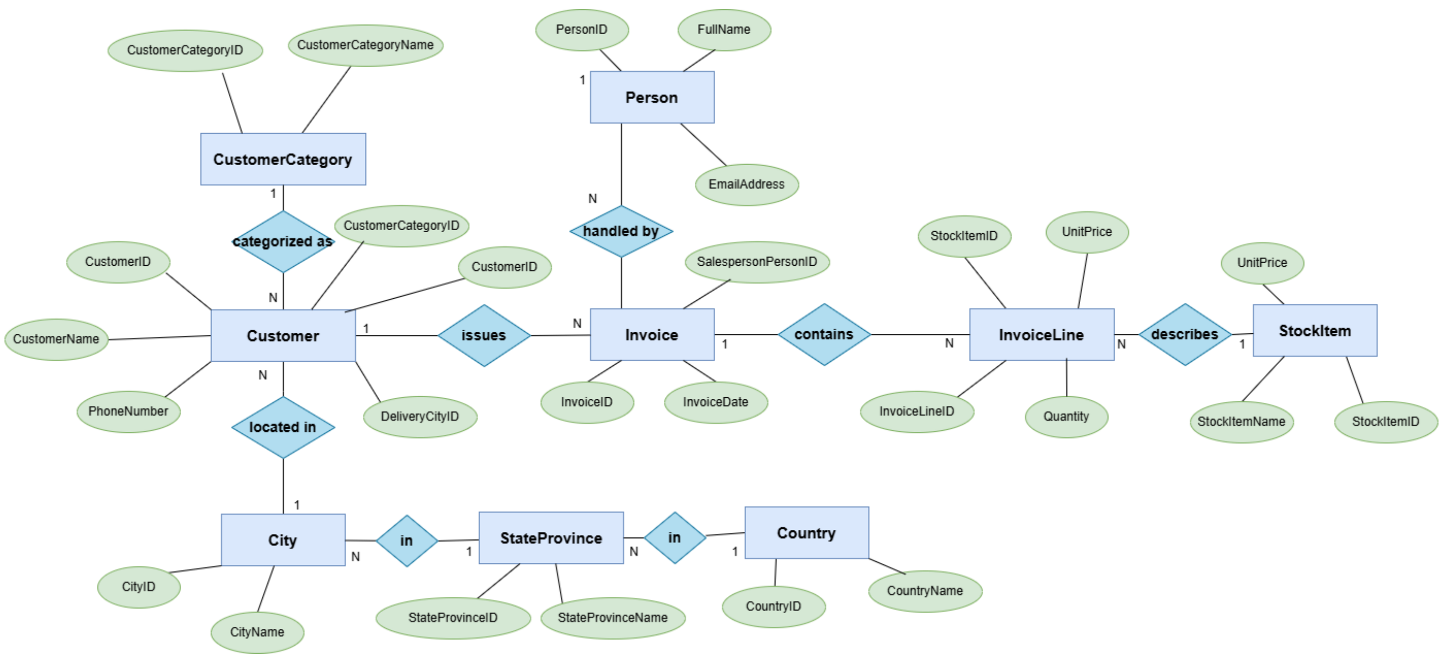
CSV File	Customer.csv	663

2.4 ER Diagram

An Entity-Relationship (ER) diagram is used to illustrate the relationships between the selected OLTP tables.

Key relationships include:

- Customers are linked to Invoices via CustomerID
- Invoices are linked to InvoiceLines via InvoiceID
- InvoiceLines are linked to StockItems via StockItemID
- Customers are linked to Cities, which are further linked to StateProvinces and Countries



3. Preparation of Data Sources

3.1 Data Source Types

Two different data source types were used in this project:

SQL Server Database - Contains core transactional data such as customers types, invoices, and products

- WideWorldImporters transactional database
- Used for extracting structured business data

CSV File - Contains customer attributes

- External customer enrichment data (customers.csv)
- Used to enhance customer dimension attributes

The SQL Server database provides structured and relational data, while the CSV file introduces additional attributes for enrichment.

3.2 Data Source Mapping

WideWorldImporters transactional database

The geographic dimension (DimLocation) contains a natural **five-level hierarchy**:

Continent → Region → Country → State province → City

Data Element	**Source**
Customer details	CSV file
Customer category	SQL Server (Sales.CustomerCategories)
Location data	SQL Server (Cities, StateProvinces, Countries)
Product details	SQL Server (StockItems)
Sales transactions	SQL Server (Invoices, InvoiceLines)

This mapping ensures that multiple data sources are integrated during the ETL process.

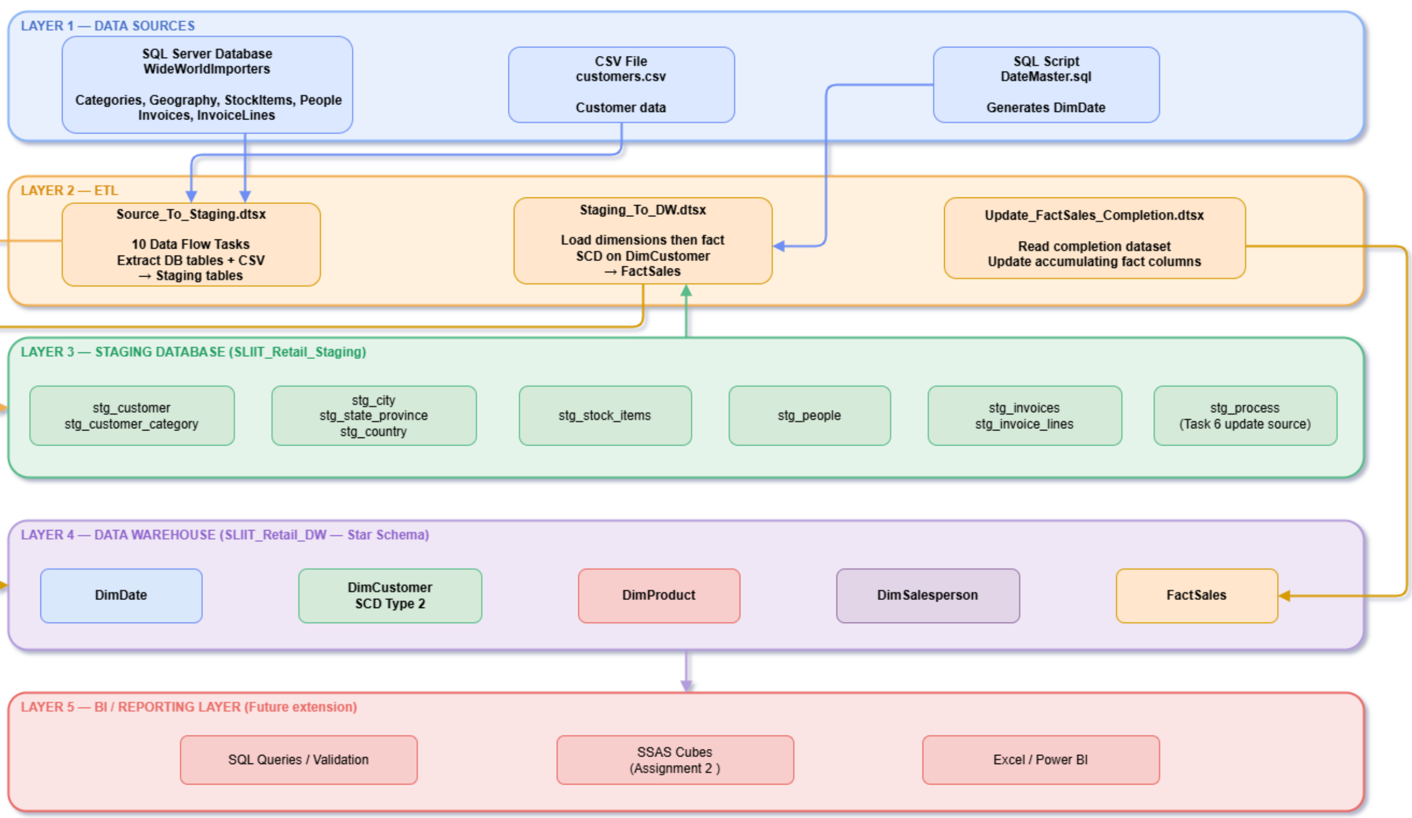
4. Solution Architecture

4.1 Architecture Diagram

The solution architecture follows a typical Data Warehouse and Business Intelligence pipeline:

Source Systems → Staging Area → Data Warehouse → Reporting

- Source Systems: SQL Server database and CSV file
- Staging Area: Temporary storage for raw data
- Data Warehouse: Dimensional model (Fact and Dimension tables)
- Reporting Layer: Used for analysis and querying



4.2 Component Description

Component	Description
Source Systems	Data extracted from SQL Server and CSV file
SSIS ETL	Performs extraction, transformation and loading
Staging Database	Stores raw data before transformation
Data Warehouse	Stores structured dimensional data
Reporting Layer	Enables analysis using queries and tools

The ETL process ensures data is cleaned, transformed, and loaded efficiently into the data warehouse.

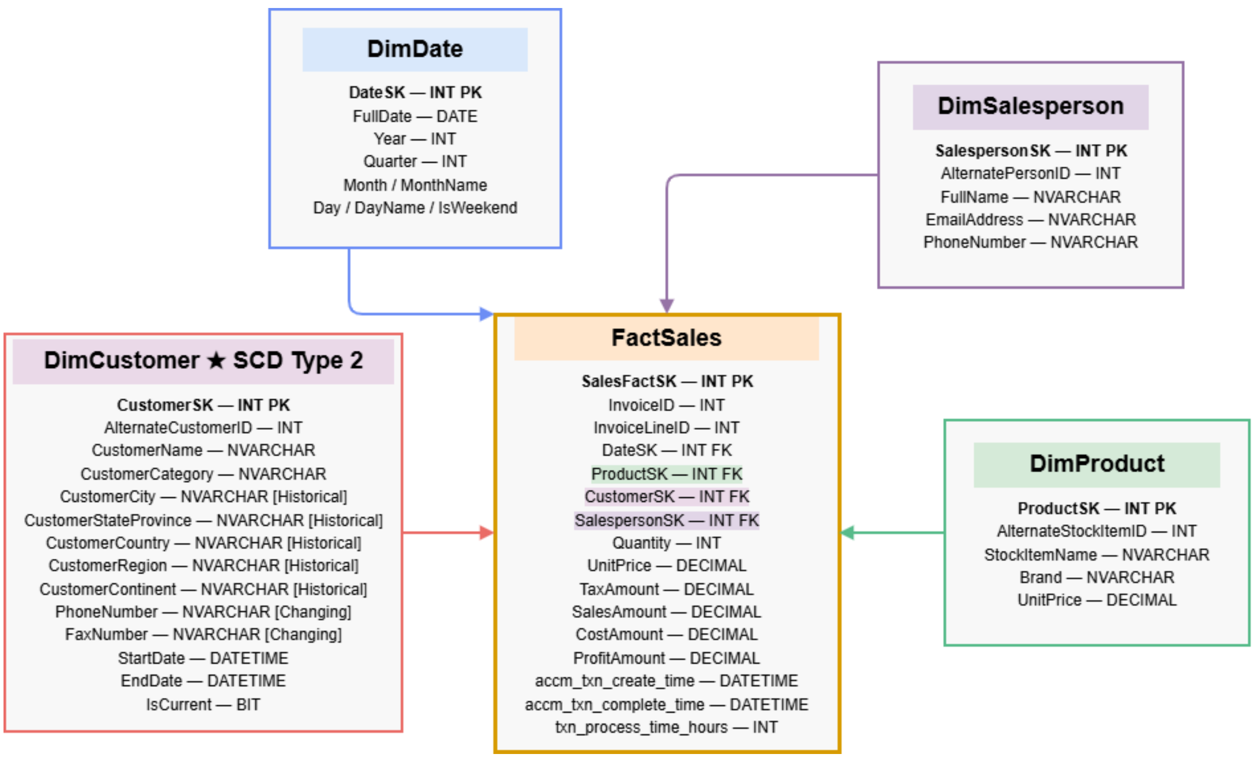
5. Data Warehouse Design and Development

5.1 Dimensional Model – Star Schema

A Star Schema design was used for the data warehouse. This model consists of one central fact table connected to multiple dimension tables.

- Fact Table: FactSales
- Dimension Tables: DimCustomer, DimProduct, DimSalesperson, DimDate

This design improves query performance and simplifies analytical processing.



The grain of the FactSales table is defined at the invoice line level. Each record represents a single product sold within a specific invoice transaction. This ensures that all measures such as sales amount, quantity, tax and profit are recorded at the most detailed level for accurate analysis.

5.2 DimDate

The DimDate table was generated using a SQL script. It contains attributes such as:

- CustomerSK
- AlternateCustomerID
- CustomerName
- CustomerCategory
- CustomerCity
- CustomerStateProvince
- CustomerCountry
- CustomerRegion
- CustomerContinent
- CustomerPhoneNumber
- CustomerFaxNumber

	DateKey	Date	FullDateUK	FullDateUSA	DayOfMonth	DaySuffix	DayName	DayOfWeekUSA	DayOfWeekUK	DayOfWeekInMonth	DayOfWeekInYear	DayOfQuarter	DayOfY
1	19900101	1990-01-01	01/01/1990	01/01/1990	1	1st	Monday	2	1	1	1	1	1
2	19900102	1990-01-02	02/01/1990	01/02/1990	2	2nd	Tuesday	3	2	1	1	1	2
3	19900103	1990-01-03	03/01/1990	01/03/1990	3	3rd	Wednesday	4	3	1	1	1	3
4	19900104	1990-01-04	04/01/1990	01/04/1990	4	4th	Thursday	5	4	1	1	1	4
5	19900105	1990-01-05	05/01/1990	01/05/1990	5	5th	Friday	6	5	1	1	1	5
6	19900106	1990-01-06	06/01/1990	01/06/1990	6	6th	Saturday	7	6	1	1	1	6
7	19900107	1990-01-07	07/01/1990	01/07/1990	7	7th	Sunday	1	7	1	1	1	7
8	19900108	1990-01-08	08/01/1990	01/08/1990	8	8th	Monday	2	1	2	2	2	8
9	19900109	1990-01-09	09/01/1990	01/09/1990	9	9th	Tuesday	3	2	2	2	2	9
10	19900110	1990-01-10	10/01/1990	01/10/1990	10	10th	Wednesday	4	3	2	2	2	10

This table supports time-based analysis and reporting.

5.3 DimCustomer

The DimCustomer table was created using

- Customer.CSV
- Sales.CustomerCategories
- Application.Cities
- Application.Countries
- Application.StateProvinces

It contains attributes such as,

- CustomerSK
- AlternateCustomerID
- CustomerName
- CustomerCategory
- CustomerCity
- CustomerStateProvince

- CustomerCountry
- CustomerRegion
- CustomerContinent
- CustomerPhoneNumber
- CustomerFaxNumber

Results Messages

	CustomerSK	AlternateCustomerID	CustomerName	CustomerCategory	CustomerCity	CustomerStateProvince	CustomerCountry	CustomerRegion	CustomerContinent	Custome
1	200	201	Tailspin Toys (Skyway	Agent	Abanda	Alabama	United States	Americas	North America	(206) 55
2	202	39	Tailspin Toys (Diablock	Agent	Abanda	Alabama	United States	Americas	North America	(270) 55
3	203	73	Tailspin Toys (Cokato	Agent	Abanda	Alabama	United States	Americas	North America	(218) 55
4	204	72	Tailspin Toys (Cortaro	Agent	Abanda	Alabama	United States	Americas	North America	(480) 55
5	205	71	Tailspin Toys (Good Hart	Agent	Abanda	Alabama	United States	Americas	North America	(231) 55
6	206	70	Tailspin Toys (New Baden	Agent	Abanda	Alabama	United States	Americas	North America	(217) 55
7	207	69	Tailspin Toys (Lytle Creek	Agent	Abanda	Alabama	United States	Americas	North America	(209) 55
8	208	68	Tailspin Toys (Gardere	Agent	Abanda	Alabama	United States	Americas	North America	(225) 55
9	209	67	Tailspin Toys (Sentinel Butte	Agent	Abanda	Alabama	United States	Americas	North America	(701) 55
10	210	66	Tailspin Toys (Madaket	Agent	Abanda	Alabama	United States	Americas	North America	(339) 55

Two view tables were created for implement the DimCustomer. For Customer, Customer.CSV and Sales.CustomerCategories were used and for Location,Cities, StateProvinces Countries tables wre used.

A Slowly Changing Dimension - SCD Type 2 approach was used to track historical changes in customer data.

5.4 DimProduct

The DimProduct table was created from,

- Warehouse.StockItems

It includes attributes such as:

- ProductSK
- AlternateStockItemID
- StockItemName
- SupplierID
- ColorID
- Brand
- Size
- UnitPrice
- TaxRate
- RecommendedRetailPrice
- TypicalWeightPerUnit
- IsChillerStock

	ProductSK	AlternateStockItemID	StockItemName	SupplierID	ColorID	Brand	Size	UnitPrice	TaxRate	RecommendedRetailPrice	TypicalWeightPerUnit
58	58	58	RC toy sedan car with remote control (Black) 1/...	10	3	Nort...	1/5...	25.00	15.000	37.38	1.500
59	59	59	RC toy sedan car with remote control (Red) 1/5...	10	28	Nort...	1/5...	25.00	15.000	37.38	1.500
60	60	60	RC toy sedan car with remote control (Blue) 1/5...	10	4	Nort...	1/5...	25.00	15.000	37.38	1.500
61	61	61	RC toy sedan car with remote control (Green) 1/...	10	NULL	Nort...	1/5...	25.00	15.000	37.38	1.500
62	62	62	RC toy sedan car with remote control (Yellow) 1...	10	36	Nort...	1/5...	25.00	15.000	37.38	1.500
63	63	63	RC toy sedan car with remote control (Pink) 1/5...	10	NULL	Nort...	1/5...	25.00	15.000	37.38	1.500
64	64	64	RC vintage American toy coupe with remote co...	10	28	Nort...	1/5...	30.00	15.000	44.85	1.700
65	65	65	RC vintage American toy coupe with remote co...	10	3	Nort...	1/5...	30.00	15.000	44.85	1.700
66	66	66	RC big wheel monster truck with remote control ...	10	3	Nort...	1/5...	45.00	15.000	67.28	1.800
67	67	67	Ride on toy sedan car (Black) 1/12 scale	10	3	Nort...	1/1...	230.00	15.000	343.85	15.000

This dimension enables product-level analysis.

5.5 DimSalesperson

The DimSalesperson table was created using:

- Application.People

It includes:

- SalespersonSK
- AlternatePersonID
- FullName
- PreferredName
- EmailAddress
- PhoneNumber
- IsEmployee
- IsSalesperson

Results | Messages

	SalespersonSK	AlternatePersonID	FullName	PreferredName	EmailAddress	PhoneNumber	IsEmployee	IsSalesperson
1	5557	1	Data Conversion Only	Data Conversion Only	NULL	NULL	0	0
2	5558	2	Kayla Woodcock	Kayla	kaylaw@wideworldimporters.com	(415) 555-0102	1	1
3	5559	3	Hudson Onslow	Hudson	hudsono@wideworldimporters.com	(415) 555-0102	1	1
4	5560	4	Isabella Rupp	Isabella	isabellar@wideworldimporters.com	(415) 555-0102	1	0
5	5561	5	Eva Muirden	Eva	evam@wideworldimporters.com	(415) 555-0102	1	0
6	5562	6	Sophia Hinton	Sophia	sophiah@wideworldimporters.com	(415) 555-0102	1	1
7	5563	7	Amy Trefl	Amy	amyt@wideworldimporters.com	(415) 555-0102	1	1
8	5564	8	Anthony Grosse	Anthony	anthonyg@wideworldimporters.com	(415) 555-0102	1	1
9	5565	9	Alica Fatnowna	Alica	alicaf@wideworldimporters.com	(415) 555-0102	1	0
10	5566	10	Stella Rosenhain	Stella	stellar@wideworldimporters.com	(415) 555-0102	1	0

This dimension supports sales performance analysis.

5.6 FactSales

The FactSales table was created using:

- Sales.Invoices
- Sales.InvoiceLines

It stores transactional data at the invoice line level.

Measures include:

- Quantity
- UnitPrice
- TaxAmount
- SalesAmount
- CostAmount
- ProfitAmount

Results Messages

	InvoiceID	InvoiceLineID	InvoiceDateKey	ConfirmedDeliveryDateKey	CustomerKey	ProductKey	SalespersonKey	Quantity	UnitPrice	TaxAmount	SalesAmount	CostAmount	ProfitAmoun
1	58898	191012	20151120	20151121	126	68	5559	5	230.00	172.50	1322.50	897.50	425.00
2	58898	191010	20151120	20151121	126	81	5559	72	18.00	194.40	1490.40	698.40	792.00
3	58899	191015	20151120	20151121	590	96	5569	84	18.00	226.80	1738.80	856.80	882.00
4	58899	191016	20151120	20151121	590	179	5569	175	1.05	27.56	211.31	123.81	87.50
5	58899	191013	20151120	20151121	590	41	5569	6	13.00	11.70	89.70	38.70	51.00
6	58899	191014	20151120	20151121	590	36	5569	6	13.00	11.70	89.70	38.70	51.00
7	58900	191017	20151120	20151121	173	84	5572	36	18.00	97.20	745.20	385.20	360.00
8	58900	191018	20151120	20151121	173	159	5572	100	18.00	270.00	2070.00	1070.00	1000.00
9	58900	191019	20151120	20151121	173	29	5572	3	13.00	5.85	44.85	19.35	25.50
10	58901	191020	20151120	20151121	123	31	5569	7	13.00	13.65	104.65	45.15	59.50

Foreign keys link to all dimension tables.

5.7 Design Assumptions

The following assumptions were made:

- Each record in the FactSales table represents a single invoice line
- Date dimension is generated manually
- Customer data is sourced from CSV
- Product and salesperson attributes are simplified for clarity
- Some attributes were excluded to maintain a simple design

6. ETL Development

The ETL (Extract, Transform, Load) process was implemented using SQL Server Integration Services (SSIS) to move and transform data from source systems into the data warehouse.

The ETL process consists of two main phases:

1. **Source to Staging**
2. **Staging to Data Warehouse**

Additionally, a separate ETL process was implemented for updating the accumulating fact table.

The ETL workflow ensures that

- Data is extracted from multiple sources (SQL Server and CSV)
- Data is transformed into a consistent format
- Data is loaded into dimension and fact tables in the correct order

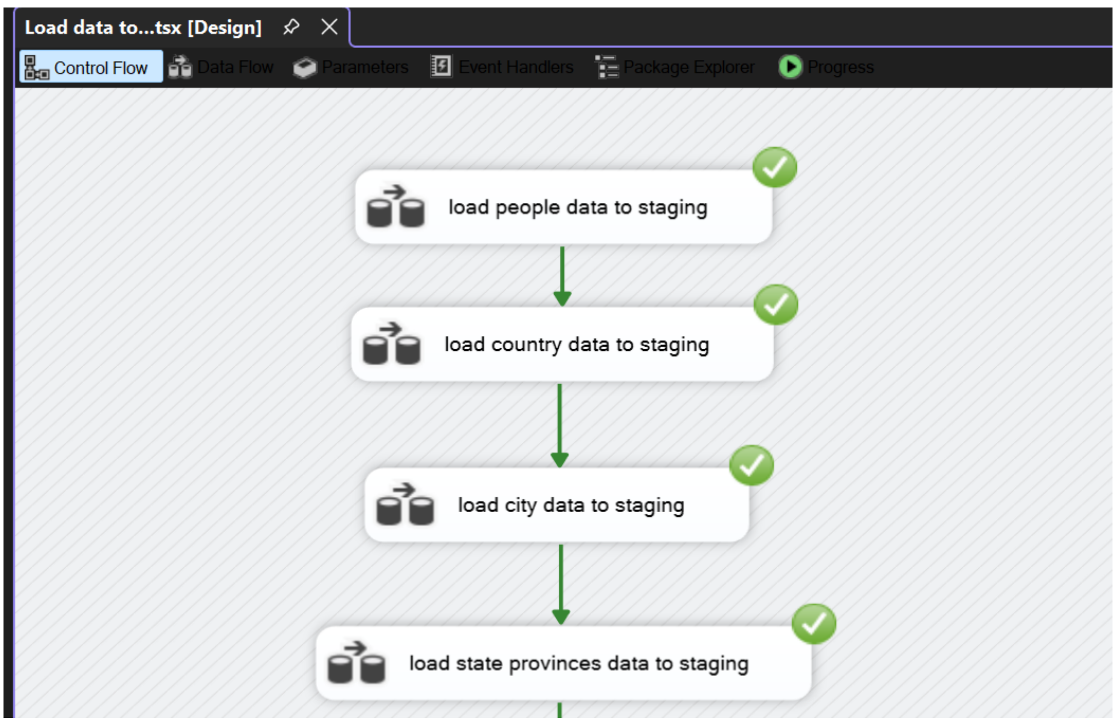
This ensures that all foreign key relationships are correctly established.

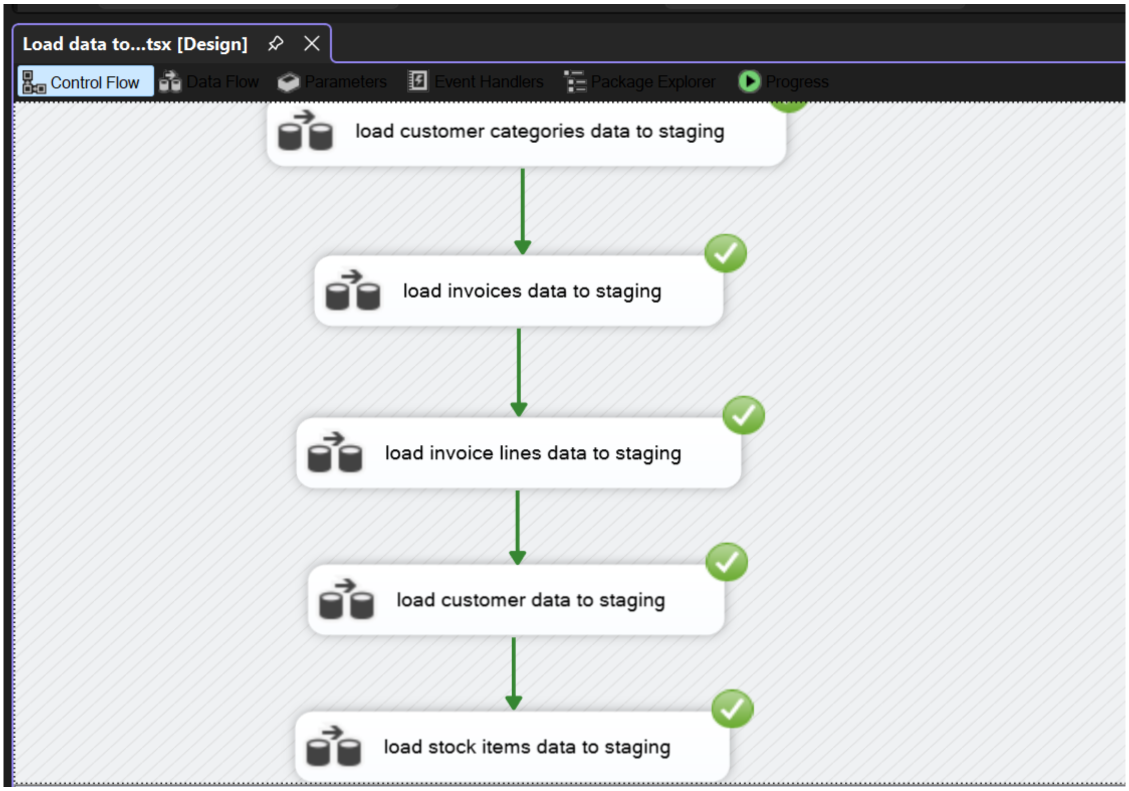
6.1 Source to Staging

- In this phase, data from source systems was extracted and loaded into staging tables using SSIS.
- The staging area acts as an intermediate layer where raw data is stored before transformation.
- The following components were used:

  - **OLE DB Source** for SQL Server tables
  - **Flat File Source** for CSV file
  - **OLE DB Destination** for loading into staging tables

Each source table was loaded into a corresponding staging table.



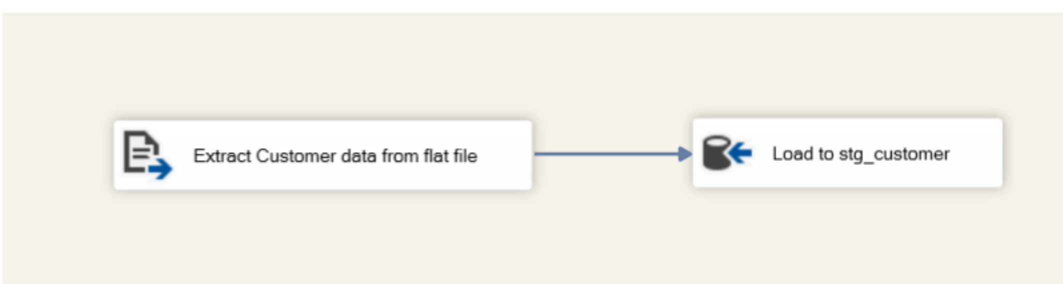


6.1.1 Load Customers Table to stg_Customers

Customer enrichment data from the CSV file was loaded using a Flat File Source.

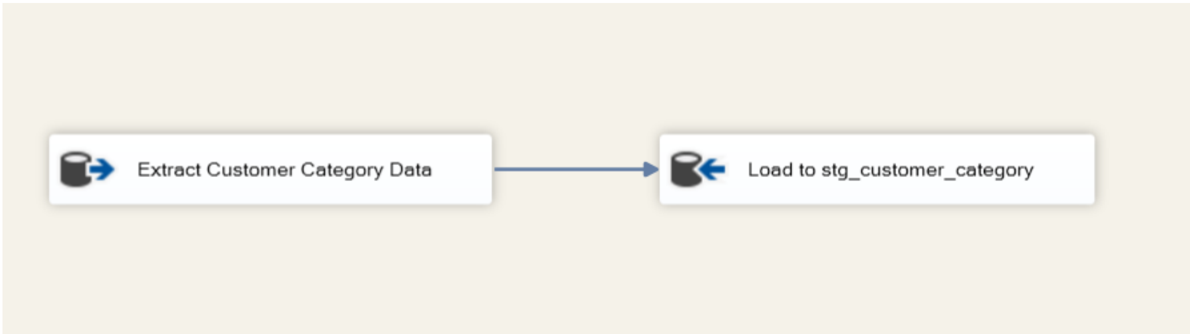
- Source - customers.csv
- Destination - stg_CustomerCSV

This data provides additional attributes such as customer segmentation.



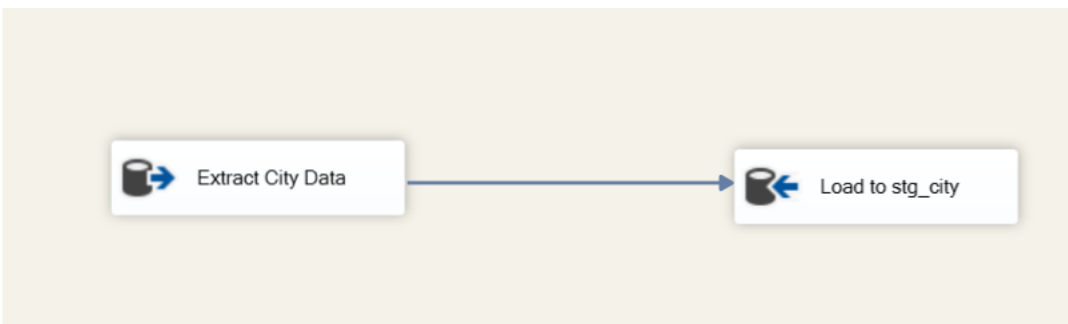
6.1.2 Load Customer Categories Table to stg_CustomerCategories

Data from Sales.CustomerCategories was loaded into stg_CustomerCategories.



6.1.3 Load Cities Table to stg_city

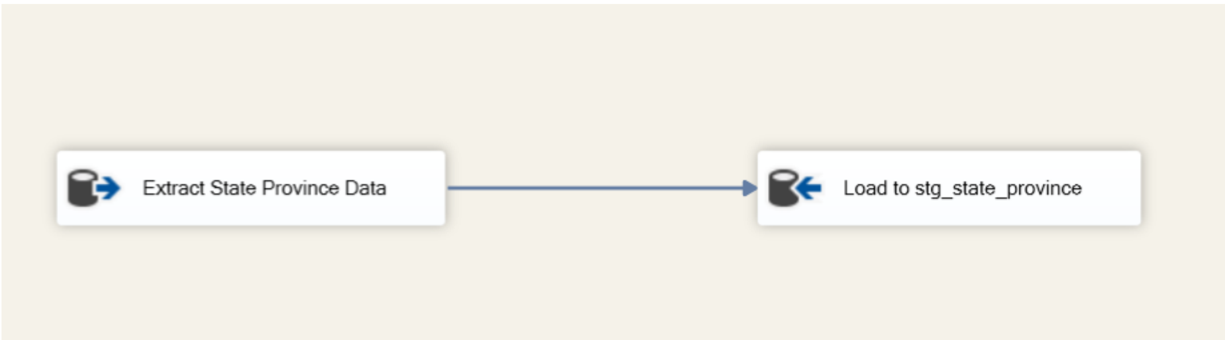
Geographical data was extracted from Application.Cities and load in to stg_Cities



These tables are used to build location hierarchy in the customer dimension.

6.1.4 Load State Provinces Table to stg_state_province

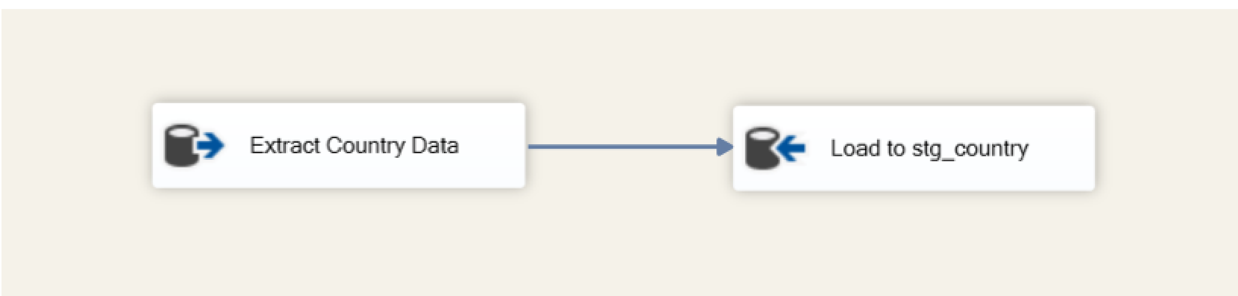
Geographical data was extracted from Application.StateProvinces and load in to stg_state_province



These table is used to build location hierarchy in the customer dimension.

6.1.5 Load Countries Table to stg_country

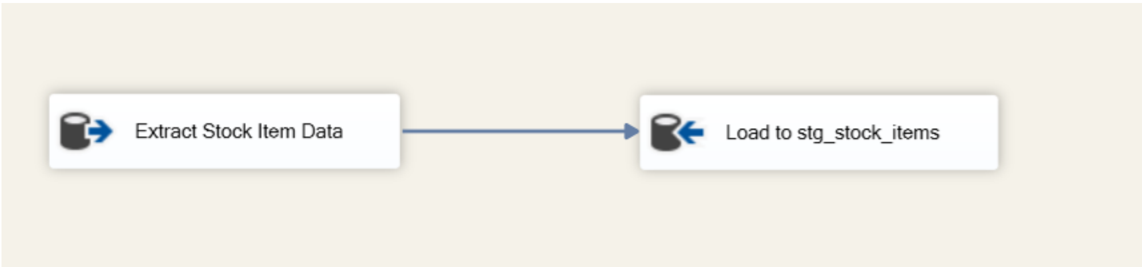
Geographical data was extracted from Application.Countries and load in to stg_country



These table is used to build location hierarchy in the customer dimension.

6.1.6 Load Stock Items Table to stg_StockItems

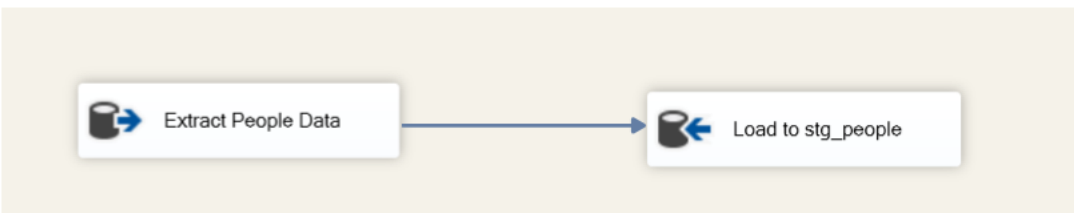
Product data from Warehouse.StockItems was loaded into stg_stock_items.



This data is later used to build the product dimension.

6.1.7 Load People Table to stg_People

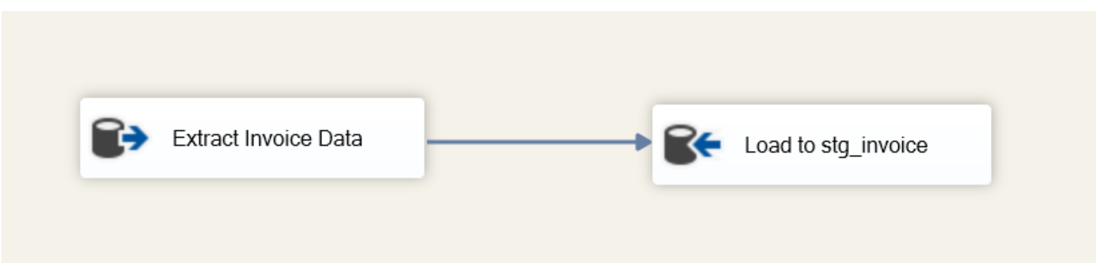
Salesperson and employee data from Application.People was loaded into stg_people.



This is used to create the salesperson dimension.

6.1.8 Load Invoices Table to stg_Invoices

Invoice header data from Sales.Invoices was loaded into stg_Invoices.



This includes customer, salesperson and date information.

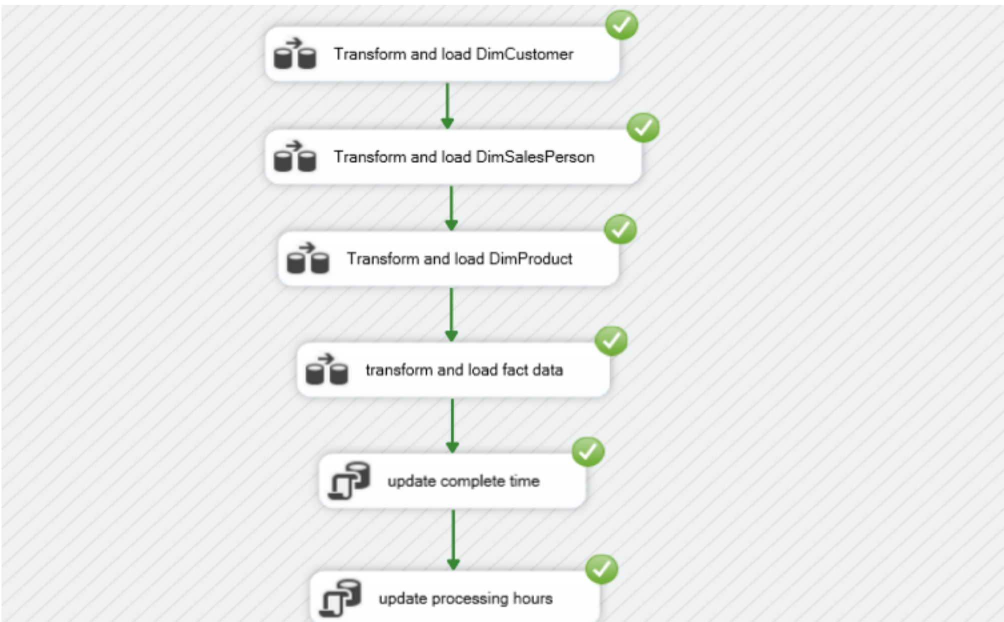
6.1.9 Load Invoice Lines Table to stg_InvoiceLines

Detailed transaction data from Sales.InvoiceLines was loaded into stg_InvoiceLines.

This table contains measures such as quantity, price and tax.

6.2 Staging to Data Warehouse

Basic error handling mechanisms were considered during ETL development. Data type conversions were validated to prevent failures during loading. Null values were handled appropriately using default values or transformations. Additionally, lookup transformations ensured referential integrity by matching surrogate keys and unmatched records were monitored to avoid data inconsistencies

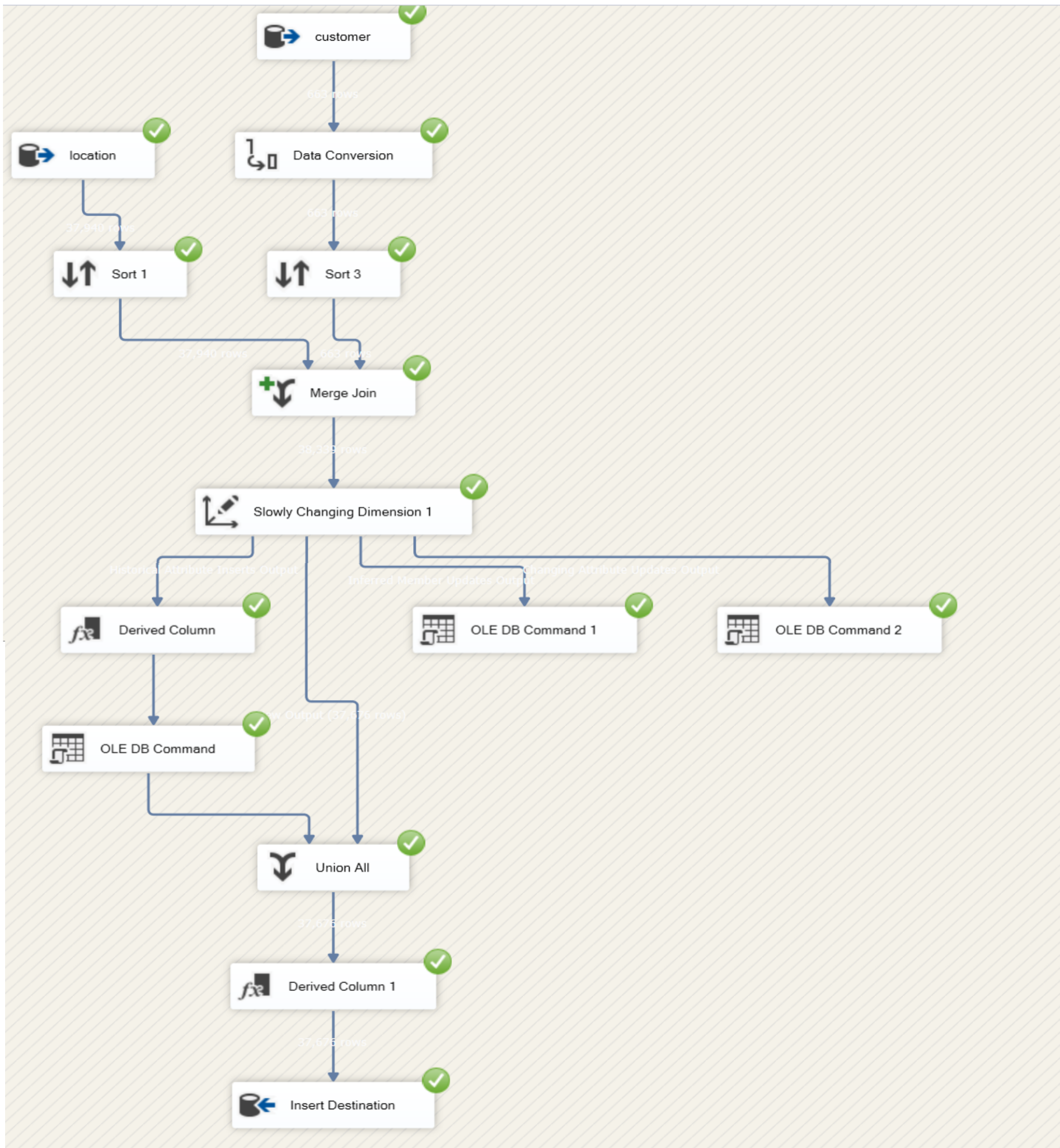


6.2.1 Load DimCustomer from Staging

The DimCustomer table was created by combining data from:

- stg_Customers
- stg_CustomerCategories
- stg_Cities
- stg_StateProvinces
- stg_Countries

Create view tables for Location by combining stg_Cities ,stg_StateProvinces, stg_Countries tables and for Customer by combining stg_Customers ,stg_CustomerCategories tables



Transformations performed:

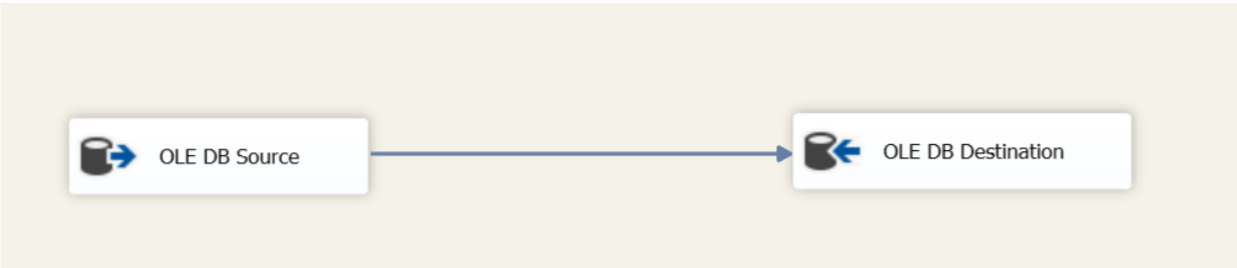
- Joining tables to enrich customer data
- Handling null values
- Applying Slowly Changing Dimension (SCD Type 2)

This allows tracking of historical changes in customer attributes.

6.2.2 Load DimProduct from Staging

The DimProduct table was populated using

- stg_StockItems



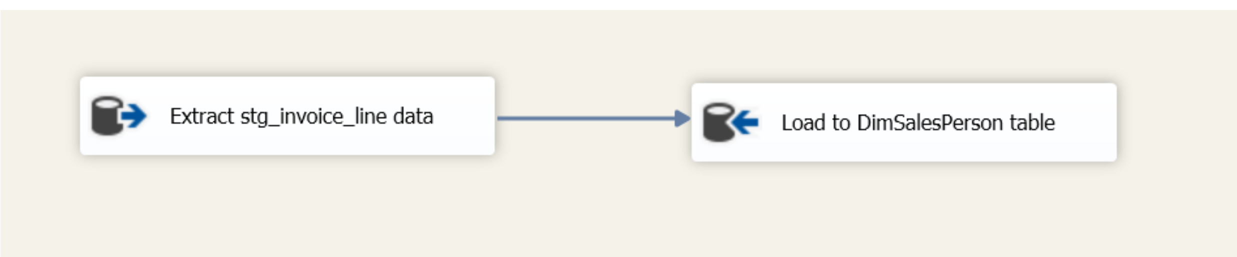
Basic transformations included

- Selecting required columns
- Renaming fields
- Removing unnecessary attributes

6.2.3 Load DimSalesperson from Staging

The DimSalesperson table was created from

- stg_People



Only records where IsSalesperson = 1 were considered.

6.2.4 Load DimDate

The DimDate table was populated using a SQL script that generates a range of dates and associated attributes.

This table is used for time-based analysis.

6.2.5 Load FactSales from Staging

The FactSales table was populated by combining

- stg_Invoices
- stg_InvoiceLines

Transformations performed

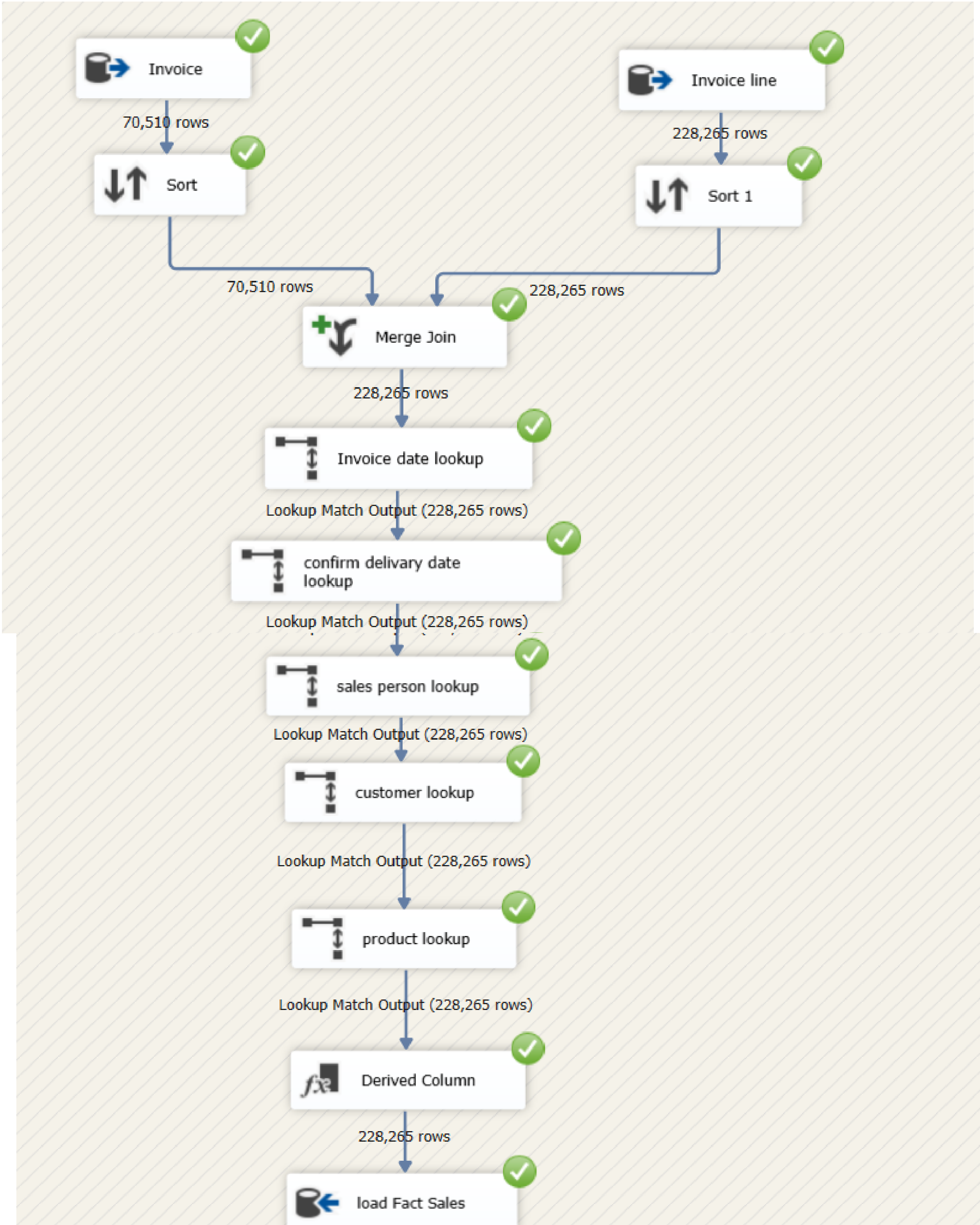
- Join between invoices and invoice lines using InvoiceID
- Lookup transformations to retrieve surrogate keys from dimension tables
- Derived columns to calculate measures

Attribute fields include

- SalesAmount
- CostAmount
- ProfitAmount

Additionally

- accm_txn_create_time was set to the current system date.


Invoice
70,510 rows
Sort
Invoice line
228,265 rows
Sort 1
70,510 rows
228,265 rows
Merge Join
228,265 rows
Invoice date lookup
Lookup Match Output (228,265 rows)
confirm delivary date lookup
Lookup Match Output (228,265 rows)
sales person lookup
Lookup Match Output (228,265 rows)
customer lookup
Lookup Match Output (228,265 rows)
product lookup
Lookup Match Output (228,265 rows)
Derived Column
228,265 rows
load Fact Sales

7. Accumulating Fact Table (Fact_Sales)

7.1 Add Accumulating Columns

The Fact_Sales table was extended with the following columns

- accm_txn_complete_time
- txn_process_time_hours

These columns track the lifecycle of a transaction.

90 % No issues found Ln: 20, Ch: 1 (491 chars, 19 lines) SPC CRLF UTF-8

Results Messages

	Price	TaxAmount	SalesAmount	CostAmount	ProfitAmount	InsertDate	ModifiedDate	accm_txn_create_time	accm_txn_complete_time	txn_process_time_hours
1	00	172.50	1322.50	897.50	425.00	2026-03-27 15:42:14.997	2026-03-27 15:42:14.997	2026-03-27 15:42:14.997	NULL	NULL
2	0	194.40	1490.40	698.40	792.00	2026-03-27 15:42:14.997	2026-03-27 15:42:14.997	2026-03-27 15:42:14.997	NULL	NULL
3	0	226.80	1738.80	856.80	882.00	2026-03-27 15:42:14.997	2026-03-27 15:42:14.997	2026-03-27 15:42:14.997	NULL	NULL
4		27.56	211.31	123.81	87.50	2026-03-27 15:42:14.997	2026-03-27 15:42:14.997	2026-03-27 15:42:14.997	NULL	NULL
5	0	11.70	89.70	38.70	51.00	2026-03-27 15:42:14.997	2026-03-27 15:42:14.997	2026-03-27 15:42:14.997	NULL	NULL
6	0	11.70	89.70	38.70	51.00	2026-03-27 15:42:14.997	2026-03-27 15:42:14.997	2026-03-27 15:42:14.997	NULL	NULL
7	0	97.20	745.20	385.20	360.00	2026-03-27 15:42:14.997	2026-03-27 15:42:14.997	2026-03-27 15:42:14.997	NULL	NULL
8	0	270.00	2070.00	1070.00	1000.00	2026-03-27 15:42:14.997	2026-03-27 15:42:14.997	2026-03-27 15:42:14.997	NULL	NULL
9	0	5.85	44.85	19.35	25.50	2026-03-27 15:42:14.997	2026-03-27 15:42:14.997	2026-03-27 15:42:14.997	NULL	NULL
10	0	13.65	104.65	45.15	59.50	2026-03-27 15:42:14.997	2026-03-27 15:42:14.997	2026-03-27 15:42:14.997	NULL	NULL

7.2 Load Completion Dataset to Staging

A separate dataset containing transaction completion times was prepared.

Structure:

- InvoiceLineID
- accm_txn_complete_time

This dataset was loaded into staging using SSIS.

Results Messages

	InvoiceLineID	accm_txn_complete_time
1	1	2026-03-26 08:00:00.000
2	2	2026-03-27 11:17:00.000
3	3	2026-03-28 14:34:00.000
4	4	2026-03-29 08:51:00.000
5	5	2026-03-30 11:08:00.000
6	6	2026-03-31 14:25:00.000
7	7	2026-04-01 08:42:00.000
8	9	2026-03-26 11:59:00.000
9	10	2026-03-27 14:16:00.000
10	8	2026-03-28 08:33:00.000

7.3 Update accm_txn_complete_time

A separate SSIS package was created to update the Fact_Sales table.

This package:

- Reads data from the completion dataset
- Matches records using InvoiceLineID
- Updates accm_txn_complete_time

7.4 Compute txn_process_time_hours

The process time was calculated using:

DATEDIFF(HOUR, accm_txn_create_time, accm_txn_complete_time)

This value was stored in txn_process_time_hours.

Results | Messages

	?rice	TaxAmount	SalesAmount	CostAmount	ProfitAmount	InsertDate	ModifiedDate	accm_txn_create_time	accm_txn_complete_time	txn_process_time_hours
1	0	4.80	36.80	16.80	20.00	2026-03-28 11:06:05.930	2026-03-28 11:06:05.930	2026-03-28 11:06:05.930	2026-03-29 11:38:00.000	24.53
2	0	259.20	1987.20	1699.20	288.00	2026-03-28 11:06:05.930	2026-03-28 11:06:05.930	2026-03-28 11:06:05.930	2026-03-30 14:55:00.000	51.82
3	00	299.25	2294.25	1384.25	910.00	2026-03-28 11:06:05.930	2026-03-28 11:06:05.930	2026-03-28 11:06:05.930	2026-03-31 08:12:00.000	69.10
4	0	3.75	28.75	22.75	6.00	2026-03-28 11:06:05.930	2026-03-28 11:06:05.930	2026-03-28 11:06:05.930	2026-04-01 11:29:00.000	96.38
5	0	9.75	74.75	32.25	42.50	2026-03-28 11:06:05.930	2026-03-28 11:06:05.930	2026-03-28 11:06:05.930	2026-03-26 14:46:00.000	-44.33
6	0	28.80	220.80	100.80	120.00	2026-03-28 11:06:05.930	2026-03-28 11:06:05.930	2026-03-28 11:06:05.930	2026-03-27 08:03:00.000	-27.05
7	0	19.20	147.20	51.20	96.00	2026-03-28 11:06:05.930	2026-03-28 11:06:05.930	2026-03-28 11:06:05.930	2026-03-28 11:20:00.000	0.23
8	0	11.70	89.70	38.70	51.00	2026-03-28 11:06:05.930	2026-03-28 11:06:05.930	2026-03-28 11:06:05.930	2026-03-29 14:37:00.000	27.52
9		79.92	612.72	267.12	345.60	2026-03-28 11:06:05.930	2026-03-28 11:06:05.930	2026-03-28 11:06:05.930	2026-03-30 08:54:00.000	45.80
10	0	26.25	201.25	113.75	87.50	2026-03-28 11:06:05.930	2026-03-28 11:06:05.930	2026-03-28 11:06:05.930	2026-03-31 11:11:00.000	72.08

7.5 Validation of Fact Table Updates

During the update process, records in the completion dataset that did not match any InvoiceLineID in the Fact_Sales table were ignored to maintain data integrity. This ensured that only valid transactions were updated, preventing incorrect or orphan updates in the fact table.

Validation queries were executed to ensure correctness

SELECT *
FROM Fact_Sales;

The results confirmed that

- Completion times were updated correctly
- Process hours were calculated accurately

	InvoiceID	InvoiceLineID	InvoiceDateKey	ConfirmedDeliveryDateKey	CustomerKey	ProductKey	SalespersonKey	Quantity	UnitPrice	TaxAmount	SalesAmount	CostAmount	ProfitAmoun
1	29876	97112	20140718	20140719	63	9	5559	1	32.00	4.80	36.80	16.80	20.00
2	29877	97116	20140718	20140719	184	149	5571	96	18.00	259.20	1987.20	1699.20	288.00
3	29877	97117	20140718	20140719	184	73	5571	7	285.00	299.25	2294.25	1384.25	910.00
4	29878	97118	20140718	20140719	572	110	5563	1	25.00	3.75	28.75	22.75	6.00
5	29878	97119	20140718	20140719	572	33	5563	5	13.00	9.75	74.75	32.25	42.50
6	29878	97120	20140718	20140719	572	5	5563	6	32.00	28.80	220.80	100.80	120.00
7	29879	97122	20140718	20140719	590	126	5558	4	32.00	19.20	147.20	51.20	96.00
8	29879	97121	20140718	20140719	590	54	5558	6	13.00	11.70	89.70	38.70	51.00
9	29880	97124	20140718	20140719	636	195	5570	144	3.70	79.92	612.72	267.12	345.60
10	29880	97123	20140718	20140719	636	62	5570	7	25.00	26.25	201.25	113.75	87.50

	Price	TaxAmount	SalesAmount	CostAmount	ProfitAmount	InsertDate	ModifiedDate	accm_txn_create_time	accm_txn_complete_time	txn_process_time_hours
1	0	4.80	36.80	16.80	20.00	2026-03-28 11:06:05.930	2026-03-28 11:06:05.930	2026-03-28 11:06:05.930	2026-03-29 11:38:00.000	24.53
2	0	259.20	1987.20	1699.20	288.00	2026-03-28 11:06:05.930	2026-03-28 11:06:05.930	2026-03-28 11:06:05.930	2026-03-30 14:55:00.000	51.82
3	00	299.25	2294.25	1384.25	910.00	2026-03-28 11:06:05.930	2026-03-28 11:06:05.930	2026-03-28 11:06:05.930	2026-03-31 08:12:00.000	69.10
4	0	3.75	28.75	22.75	6.00	2026-03-28 11:06:05.930	2026-03-28 11:06:05.930	2026-03-28 11:06:05.930	2026-04-01 11:29:00.000	96.38
5	0	9.75	74.75	32.25	42.50	2026-03-28 11:06:05.930	2026-03-28 11:06:05.930	2026-03-28 11:06:05.930	2026-03-26 14:46:00.000	-44.33
6	0	28.80	220.80	100.80	120.00	2026-03-28 11:06:05.930	2026-03-28 11:06:05.930	2026-03-28 11:06:05.930	2026-03-27 08:03:00.000	-27.05
7	0	19.20	147.20	51.20	96.00	2026-03-28 11:06:05.930	2026-03-28 11:06:05.930	2026-03-28 11:06:05.930	2026-03-28 11:20:00.000	0.23
8	0	11.70	89.70	38.70	51.00	2026-03-28 11:06:05.930	2026-03-28 11:06:05.930	2026-03-28 11:06:05.930	2026-03-29 14:37:00.000	27.52
9		79.92	612.72	267.12	345.60	2026-03-28 11:06:05.930	2026-03-28 11:06:05.930	2026-03-28 11:06:05.930	2026-03-30 08:54:00.000	45.80
10	0	26.25	201.25	113.75	87.50	2026-03-28 11:06:05.930	2026-03-28 11:06:05.930	2026-03-28 11:06:05.930	2026-03-31 11:11:00.000	72.08

8. Conclusion

This project successfully demonstrated the end-to-end development of a data warehouse solution using the WideWorldImporters dataset. A dimensional model was designed using a star schema, with clearly defined fact and dimension tables to support efficient analytical processing. The implementation of the ETL process ensured that data was accurately extracted, transformed and loaded from the OLTP system into the data warehouse while maintaining data quality and consistency.

Additionally, the creation of an accumulating fact table enabled tracking of transaction lifecycle and processing time, providing deeper analytical insights. The development of an OLAP cube further enhanced the system by allowing multidimensional analysis and supporting complex queries for decision-making.

Overall, the solution provides a scalable and efficient platform for business intelligence and reporting. Future improvements could include the integration of real-time data processing and advanced analytics techniques to further enhance the capabilities of the system.